MiPhyllo: taxon-stratified 16S and 28S PCR primer candidates for environmental DNA metabarcoding of Phyllodocida

Fei Xia

Computational preprint · 5 November 2026

Abstract

Taxonomic breadth and primer specificity can conflict when designing environmental DNA assays for diverse marine annelids. We integrated mitochondrial 16S and nuclear 28S designs across eight nested target units within Phyllodocida: the order, four suborders and three families. Frozen, species-deduplicated panels were searched for short, medium and long windows. Ordinary and degenerate counterparts shared binding endpoints, with degeneracy limited to eight variants per oligo. The catalogue comprises 94 current pairs and eight conditional alternatives. All 102 pairs were re-scanned using independent C++ and Python implementations with agreement in target and background outcomes. Family-level designs improved focal coverage: Nereididae 16S LD covered 58/61 references, while Hesionidae 28S SD covered 40/45. Order-wide coverage remained low. Latest Nereididae 28S SD covered 29/52 references with eight exact matches. A conditional Syllidae 16S S2D increased coverage from 110 to 126/180 and exact matches from seven to 48, but exhibited substantially greater loose-model background matching. These results support taxon-stratified candidate selection rather than a single universal assay. They represent computational matching, not experimentally measured amplification performance.

1. Introduction

Phyllodocida comprises divergent polychaete lineages for which a single short rRNA assay may not reconcile coverage and specificity. Scale-worm phylogenetic reference data provide an established starting point [1], while environmental DNA has been applied to marine metazoan surveys [2]. Broad-target primer designs such as MiFish illustrate the value of conserved binding regions surrounding variable sequence [3]; this study applies that general principle to annelid reference panels, without transferring fish-primer sequences or assuming equivalent performance.

We compare designs at the order, suborder and family levels, retain multiple amplicon lengths, and explicitly separate strict and loose background matching. MiPhyllo denotes the integrated Phyllodocida collection. The repository address and MiAphro sequence identifiers remain stable for traceability; the original Aphroditiformia preprint remains a separate case study.

2. Materials and methods

2.1 Frozen references and taxonomic units

Each gene panel retained one reference per named species, prioritizing sequence completeness during source curation. Target units were Phyllodocida; Aphroditiformia, Glyceriformia, Nereidiformia and Phyllodociformia; and Syllidae, Nereididae and Hesionidae. Panels overlap taxonomically and must not be summed as independent sampling. Direction-adjusted MAFFT alignments [4] guided homologous windows. Reference accessions, family assignments, alignments and frozen backgrounds are distributed with each unit.

2.2 Candidate generation

Current selected spans fall within S 100–200 bp, M 201–300 bp and L 301–480 bp. Earlier searches used narrower ranges, retained in source scripts. Latest expanded searches evaluated 18–30 nt oligos at 1 nt start increments, combining frequent haplotypes and consensus seeds, coverage-oriented and exact-oriented branches, spatially distributed windows and local refinement. Ordinary/D pairs share four binding endpoints; each oligo has degeneracy ≤8, restricted to its 3′ half. An ordinary sequence need not be nested within the D expansion.

Two concentration scenarios—50 nM per variant and 200 nM per direction—required minimum predicted Tm ≥45°C, F/R midpoint difference ≤5°C and predicted secondary-structure Tm <55°C. Search procedures were staged heuristics, not exhaustive enumeration of all possible sequences.

2.3 Matching and selection

Target coverage required ≤2 mismatches per end, compatibility at each terminal three bases, total product length 100–500 bp and internal length ≥50 bp. Exact matching required zero mismatches at both ends, including IUPAC-compatible bases. All reference species remained in the denominator, including fragments lacking a complete window. Background scans allowed an internal gap of 0–1000 bp. Strict WC/GT used ≤2 mismatches and terminal compatibility; loose WC allowed ≤3 mismatches and loose GT ≤2 without terminal constraints. GT additionally permits primer-oriented G:A and T:C compatibility and is a sensitivity model.

The original Aphroditiformia study used a six-group polychaete background; Phyllodocida used an expanded marine background. Both retained absolute limits of ten strict and fifty loose hits per model. Later suborder/family designs required strict WC ≤1%, GT ≤3%, loose WC ≤5% and GT ≤10%, plus each strict close-relative count ≤max(2, floor(3% × close-relative N)). Marine screening was based on existing genus-level WoRMS assignments, not comprehensive species/locality verification.

Latest balanced selections did not reduce ordinary/D coverage or exact matching relative to preceding selections. Conditional alternatives retain strict and close-relative gates but remove loose-model hard vetoes. They are separate candidates, identified by 2 (e.g., S2/S2D), not two-pair mixtures.

2.4 Independent verification

All 102 selected pairs were freshly scanned on their own frozen panels, comparing C++ and independent Python outcomes record by record. Sequence exports, coordinates, degeneracy, species totals, bilingual tables and checksums were also checked. No wet-lab validation, common-panel comparison or independent holdout experiment is reported here.

3. Results

3.1 Reference panels and coverage

Target unit	16S target	16S background	28S target	28S background
Aphroditiformia	94	381	68	214
Phyllodocida	515	3340	302	2085
Glyceriformia	20	3835	12	2375
Nereidiformia	315	3540	133	2254
Phyllodociformia	59	3796	63	2324
Syllidae	180	3675	28	2359
Nereididae	61	3794	52	2335
Hesionidae	53	3802	45	2342

Table 1. Frozen reference species counts. Nested panels overlap; background definitions differ between design units.

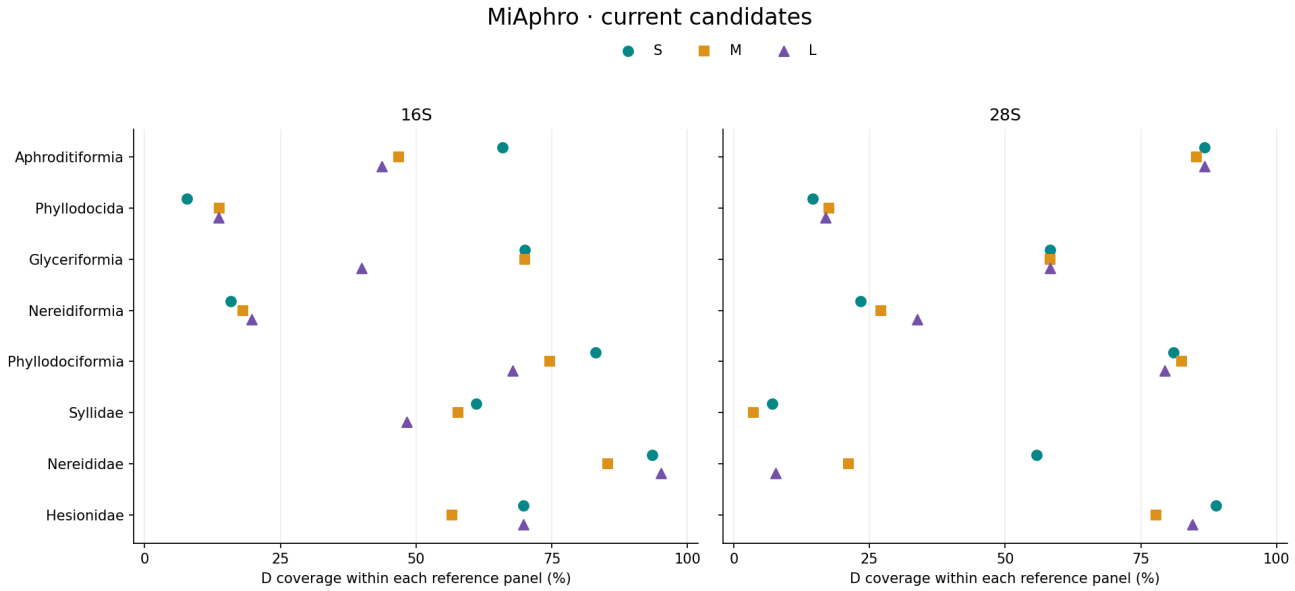


Figure 1. Current D coverage within each reference panel. S, M and L denote selected length classes. Missing Syllidae 28S L indicates no passing pair. Percentages are not common-panel performance comparisons.

Phyllodocida designs covered at most 71/515 16S and 53/302 28S references. Nereidiformia remained limited (62/315 and 45/133). Phyllodociformia reached 49/59 and 52/63, but its references represented Phyllodocidae only. Glyceriformia 16S SD/MD covered 14/20, with only one exact match each. Detailed ordinary/D outcomes and sequences for all units appear in Appendix A.

3.2 Family-level optimization and conditional alternatives

Nereididae 16S LD covered 58/61 references, but its strict WC/GT background counts (30/58) exceeded those of SD (0/12), which covered 57/61. Latest 28S SD increased coverage from 28 to 29/52 while retaining eight exact matches; all four strict/loose background counts were zero. The 28S MD and LD retained low current coverage (11/52 and 4/52).

Syllidae current 16S SD, MD and LD covered 110, 104 and 87 of 180 references, respectively. No ordinary/D replacement improved or retained all four target metrics under the original complete background profile. Conditional S2D covered 126/180 with 48 exact matches, compared with 110 and seven for SD. However, strict WC/GT hits were 33/42 of 3,675 background references and loose hits were 1,007/601. Thus the apparent gain depends on how mismatch tolerance is treated. The conditional set is not merged into the current FASTA.

4. Discussion

The data favor stratified target design: broad order-level candidates show poor reference coverage, whereas several narrower panels produce higher within-panel coverage. This pattern does not establish superiority in a common sample or reference panel. Some gains reflect restricted target scope, different reference availability and a different near-relative background.

Degeneracy can increase compatible target diversity without increasing the number of primer pairs, but exact matching, background compatibility and the concentration of each variant remain distinct considerations. A high exact-match count does not rescue a low-coverage design, and low strict-background counts do not imply low loose-model matching. The conditional Syllidae designs make this tradeoff explicit.

Reference limitations include incomplete fragments, uneven taxonomic representation and inherited taxonomy. Lacydoniidae is absent from Glyceriformia references, Microphthalmidae from Nereidiformia references, and Phyllodociformia includes only Phyllodocidae. Syllidae 28S remains poorly covered. Neither all described species nor the full marine environmental community is represented. Primer-site matching also does not measure taxonomic resolution of the intervening amplicon.

5. Conclusion and data availability

MiPhyllo provides a traceable, taxon-stratified collection of 16S/28S candidates and explicit alternatives for eight Phyllodocida target units. Candidate use should consider target scope, exact matching and both background models. Frozen references, 94 current pairs, eight conditional pairs, per-species outcomes, source algorithms and independent re-scan records are available at github.com/GUIBA-EX/MiAphro. Full canonical identifiers follow MiAphro-{target}-{gene}-{length}[2][D]_{F|R}; these stable identifiers are retained despite the expanded manuscript name.

References

[1] Norlinder E et al. (2012). Phylogeny of scale-worms (Aphroditiformia, Annelida), assessed from 18SrRNA, 28SrRNA, 16SrRNA, mitochondrial cytochrome c oxidase subunit I (COI), and morphology. *Molecular Phylogenetics and Evolution*. PMID: 22789762.

[2] Environmental DNA metabarcoding for monitoring metazoan biodiversity in Antarctic nearshore ecosystems (2021). *PeerJ* 9:e12458. doi:10.7717/peerj.12458.

[3] Miya M et al. (2015). MiFish, a set of universal PCR primers for metabarcoding environmental DNA from fishes: detection of more than 230 subtropical marine species. *Royal Society Open Science* 2:150088. doi:10.1098/rsos.150088.

[4] Katoh K, Standley DM (2013). MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Molecular Biology and Evolution* 30:772–780. doi:10.1093/molbev/mst010.

Appendix A1. Aphroditiformia

Canonical prefix: MiAphro-Aphroditiformia-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	138	58/94	1/94	0/0	23/0
16S-SD	138	62/94	1/94	0/0	25/0
16S-M	210	28/94	24/94	0/0	0/0
16S-MD	210	44/94	24/94	2/9	12/9
16S-L	329	28/94	24/94	0/0	0/0
16S-LD	329	41/94	24/94	0/7	7/7
28S-S	148	54/68	29/68	0/0	16/2
28S-SD	148	59/68	47/68	2/2	19/2
28S-M	250	56/68	50/68	0/0	16/0
28S-MD	250	58/68	53/68	1/2	17/2
28S-L	368	58/68	22/68	0/0	15/0
28S-LD	368	59/68	33/68	0/1	18/1

Primer sequences

Gene/pair	Forward	Reverse
16S-S	ATAAACTACCTTAGGGATAACA	GAGATTTAACAGTTGAACAAAC
16S-SD	ATAAACTACCTYRGGGATAACA	GAGATTTAACAGTTGAACRAAC
16S-M	AAGACAAGAAGACCCTATAG	ACCATCCTATCAATAAGAGCTC
16S-MD	AAGACAAGAAGACCCYRTWG	ACCATCCTATCAATWWGARCTC
16S-L	GTAGCATAATCATTTGCCTT	ACCATCCTATCAATAAGAGC
16S-LD	GTAGCATAATMAYTTGCCTY	ACCATCCTATCAATWWGARC
28S-S	CTCAGAGGCAAACTGATGGGAC	GCTCTCCTCGTGGAAAGGGC
28S-SD	CTCAGAGGCAAACYGATGGGAC	GCTCTCCTCGYRGAAAGKGC
28S-M	AATCCCCTGTCGCTGCGGCA	TCCCTCACGGTACTTGTCCGCT
28S-MD	AATCCCCTGTCGYTGCGGCA	TCCCTCACGGTACTTGTCCKCT
28S-L	CTCAGAGGCAAACTGATGGGAC	CGGAACACACTGTAAACAGTCC
28S-LD	CTCAGAGGCAAACYGATGGGAC	CGGAACACACTGTRAACAGTCC

Binding coordinates, accession identifiers and per-species matches: panels/Aphroditiformia/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A2. Phyllodocida

Canonical prefix: MiAphro-Phyllodocida-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	178	36/515	13/515	0/3	0/3
16S-SD	178	40/515	16/515	0/6	0/6
16S-M	256	69/515	10/515	2/5	34/7
16S-MD	256	71/515	10/515	3/6	36/8
16S-L	380	69/515	14/515	3/4	30/5
16S-LD	380	70/515	14/515	3/4	33/5
28S-S	136	39/302	16/302	0/0	0/0
28S-SD	136	44/302	24/302	0/0	0/0
28S-M	262	45/302	15/302	1/2	4/3
28S-MD	262	53/302	22/302	4/6	15/9
28S-L	320	46/302	26/302	0/0	0/0
28S-LD	320	51/302	38/302	0/0	4/0

Primer sequences

Gene/pair	Forward	Reverse
16S-S	AGGACACACAAGAATAAAGCTG	TGTCTTTTGGTTGTCCCAAC
16S-SD	AGGACACACAAGRATWAARCTG	TGTCTTTTGGTYGYCCCARC
16S-M	AAGGACAAGAAGACCCTATAGA	ATAGTTACACTAAGCCAACATC
16S-MD	AAGGACAAGAAGACCCTRTNGA	ATAGTTACACTRAGCCAACATC
16S-L	CAAAGGTAGCATAATCATTTGC	ATAGTTACACTAAGCCAACATC
16S-LD	CAAAGGTAGCAYAATMAYTTGC	ATAGTTACACTRAGCCAACATC
28S-S	GTGTTATAGTCCCTGGGTTC	GGAACACACTGTAAACAGTCCA
28S-SD	GTGTTATAGTCCCYGGKKTC	GGAACACACTGTRRACAGTCCW
28S-M	AGAGGCAAACGGATAGGACC	GCCACCCGAGGCTATAACAC
28S-MD	AGAGGCAAACKGAYRGGACC	GCCACCCGAGTCTATAMYRC
28S-L	GGAGGGCGCCCTTTCCACGAGG	CGATCGATTCGCCACGGACCCT
28S-LD	GGAGGGCGCCCTTTCCRYGAGS	CGATCGATTCGCCASNGACCCT

Binding coordinates, accession identifiers and per-species matches: panels/Phyllodocida/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A3. Glyceriformia

Canonical prefix: MiAphro-Glyceriformia-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	110	8/20	0/20	2/3	60/27
16S-SD	110	14/20	1/20	14/33	113/104
16S-M	252	7/20	1/20	1/1	40/17
16S-MD	252	14/20	1/20	10/13	81/35
16S-L	316	5/20	1/20	0/0	2/0
16S-LD	316	8/20	2/20	0/7	10/8
28S-S	106	7/12	4/12	0/2	1/28
28S-SD	106	7/12	6/12	0/3	7/31
28S-M	236	5/12	4/12	0/4	90/35
28S-MD	236	7/12	4/12	0/4	93/35
28S-L	324	5/12	4/12	0/1	49/26
28S-LD	324	7/12	4/12	0/1	49/26

Primer sequences

Gene/pair	Forward	Reverse
16S-S	TAAATTTTGTTGGGGCGACA	ATCCCTAAGGTAGCTATTTCCT
16S-SD	TAAATTTTGTTGGGGCGACA	ATCCCTATGGTAGCTRWTTYCT
16S-M	CACGAAAATAAAGCTGTCTC	CCCTAAGGTAGCTATTTCCT
16S-MD	CACGAAAATAAWDCTGTCTC	CCCTAAGGTAGCTRTTTYCT
16S-L	AACTAGACTTCAAGTGAA	AACCTTTCTAGGCTTCTG
16S-LD	AACTAGACCYYARGTGAA	AACCTTTCTWRGCTTCTR
28S-S	ACGGCGAGTGAAGCGGGAAGAG	CTCGGACGCCAGAACGCTGC
28S-SD	ACGGCGAGTGAAGCGGGAWGAG	CTCGGACGCCASAMCRCTGC
28S-M	GCGGACAAGTACCGTGAG	GAAAACCGGTCGTGGCTA
28S-MD	GCGGACAAGTACCGTGAG	GAAAACCGGTCGTGGCYA
28S-L	CGTCCTTGGAGTCGGGTTGT	CGGAAAACCGGTCGTGGCTA
28S-LD	CGTCCTTGGAGTCGGGTTGT	CGGAAAACCGGTCGTGGCYA

Binding coordinates, accession identifiers and per-species matches: panels/Glyceriformia/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A4. Nereidiformia

Canonical prefix: MiAphro-Nereidiformia-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	162	45/315	3/315	0/0	0/0
16S-SD	162	50/315	15/315	0/0	0/0
16S-M	212	55/315	8/315	2/4	36/7
16S-MD	212	57/315	8/315	3/7	46/12
16S-L	314	48/315	3/315	1/7	30/8
16S-LD	314	62/315	0/315	15/40	74/53
28S-S	194	23/133	4/133	0/0	0/1
28S-SD	194	31/133	0/133	0/6	53/55
28S-M	252	28/133	6/133	1/4	17/6
28S-MD	252	36/133	7/133	3/6	51/9
28S-L	324	27/133	6/133	0/1	22/6
28S-LD	324	45/133	6/133	11/26	100/32

Primer sequences

Gene/pair	Forward	Reverse
16S-S	TGGGACAACCTAAAGACA	AATAGTTACACTAAGCCAACAT
16S-SD	TGGGACAACCWWAAGMCA	AATGGTTACACTAAGCCAACAT
16S-M	AAGGACAAGAAGACCCTA	AATAGTTACACTAAGCCAAC
16S-MD	AAGGACAAGAAGACCCTA	AATAGTTACAYWRAGCCAAC
16S-L	GCCTTTTAATTGGAGGCTAG	AATGGTTACACTAAGCCA
16S-LD	GCCTTTTAATTRGAGGMTWG	AATAGGTACVYTAAGCCA
28S-S	GGGACCGCAGAGTCTGCC	CCACCATCCCGCCGGCCT
28S-SD	GGGACCGCAAAGTCTRSY	CTACCATCCCSTSGGMCT
28S-M	GACGAGCGCACTTTCCGC	GAAGCACTGTGAACAGTCCT
28S-MD	GACGAGCGCACTTTCYGB	AAAGCACTGTRRACAGTCCW
28S-L	GACGCGTGCACTTTCCGG	GTGCCGACAGATTCGCCA
28S-LD	GACGCGTGCACTTTCCRB	GTGCCGACARAWTCGCYA

Binding coordinates, accession identifiers and per-species matches: panels/Nereidiformia/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A5. Phyllodociformia

Canonical prefix: MiAphro-Phyllodociformia-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	118	31/59	2/59	0/0	10/3
16S-SD	118	49/59	11/59	0/8	81/11
16S-M	258	33/59	0/59	12/49	47/57
16S-MD	258	44/59	0/59	15/93	76/101
16S-L	308	27/59	2/59	0/0	9/4
16S-LD	308	40/59	3/59	0/1	76/29
28S-S	198	49/63	4/63	5/18	35/30
28S-SD	198	51/63	4/63	7/23	53/53
28S-M	298	12/63	6/63	0/0	44/12
28S-MD	298	52/63	5/63	21/26	95/42
28S-L	302	12/63	0/63	0/0	36/11
28S-LD	302	50/63	0/63	14/20	82/35

Primer sequences

Gene/pair	Forward	Reverse
16S-S	ATAGTTACCATAGGGATAACAGGC	ACCAACCCTTATAGACTGCTAC
16S-SD	ATAGTTACCATAGGGATAACAGGC	ACCAACCCTTATARRCTGCTRC
16S-M	GACGAAAAGACCCTATTGAG	CTATAGGATATCCTAAGCCAACAT
16S-MD	GACGAAAAGACCCTATYGAK	CTATAGGATATCYTAAGCCAACAT
16S-L	ACCTAAATTTTAATGAAAGACA	TTGAACAAACCAACCCTT
16S-LD	ACCTAAATTTTAATRWARGACA	TTGAACAAACYWACCCTY
28S-S	GAAAGCCACGACCGGTTC	CACTGTGAACAGTCCTCGCCAG
28S-SD	GAAAGCCACGACCGGTTC	CACTGTGAACAGTCCTCVCCWG
28S-M	GTCCTTGGAGTCGGGTTGTTTG	AAAGTGCGCCCACCGACCAC
28S-MD	GTCCTTGGAGTCGGGTTGTTTG	AAAGTGCACCCACCGACNRC
28S-L	GCGTCCTCGGAGTCGGGT	GGAAAGTGCGCCCACCGACCAC
28S-LD	GCGTCCTCGGAGTCGGGT	GGAAAGTGCACCCACCGACNRC

Binding coordinates, accession identifiers and per-species matches: panels/Phyllodociformia/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A6. Syllidae

Canonical prefix: MiAphro-Syllidae-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	142	55/180	2/180	0/2	4/2
16S-SD	142	110/180	7/180	1/2	15/2
16S-M	282	41/180	3/180	2/4	53/4
16S-MD	282	104/180	0/180	6/23	162/24
16S-L	308	39/180	3/180	0/0	17/4
16S-LD	308	87/180	7/180	2/11	169/41
28S-S	118	2/28	2/28	0/0	0/0
28S-SD	118	2/28	2/28	0/0	0/0
28S-M	202	1/28	1/28	0/0	0/0
28S-MD	202	1/28	1/28	0/0	0/0

Primer sequences

Gene/pair	Forward	Reverse
16S-S	AAAGACAGAAAGACCCTA	CCCTACAGTAGCTTACTTAT
16S-SD	AAAGACAGAAAGWCCCTH	CCCTACAGTARCTTACTTRW
16S-M	GCGAAGGTAGCATAATCATT	AAAATATTAGCCTGTTATCCCTACAGTA
16S-MD	GCGAAGGTAGCAYAATCAYK	AAAGTATTAGCCTGTTAYCCCTAVAGTA
16S-L	TTCAACGGCCGCGGTATC	AAAGTATTAGCCTGTTATCCCTACAG
16S-LD	TTCAACGGCCGCGGTABY	AAAGTATTAGCCTGTTATCCCTANAG
28S-S	CTGTGGCGTGTGGGACGG	GGACCATAGGGAGGAGGCCGGA
28S-SD	CTGTGGCGTGTGGGACGG	GGACCATAGGGAGGAGRSCSGA
28S-M	TGCGCCGGTGGCAGCATCTG	ACTTTCCCTCACGGTACTTGTTCGCTAT
28S-MD	TGCGCCGGTGGCAGCATCTG	ACTTTCCCTCACGGTACTTGTYCRCTAT

Binding coordinates, accession identifiers and per-species matches: panels/Syllidae/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A7. Nereididae

Canonical prefix: MiAphro-Nereididae-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	194	30/61	7/61	0/0	16/6
16S-SD	194	57/61	7/61	0/12	51/25
16S-M	212	52/61	8/61	5/9	48/12
16S-MD	212	52/61	8/61	8/14	66/19
16S-L	412	38/61	6/61	0/40	66/106
16S-LD	412	58/61	11/61	30/58	166/285
28S-S	150	19/52	1/52	0/0	0/0
28S-SD	150	29/52	8/52	0/0	0/0
28S-M	202	1/52	1/52	0/0	0/0
28S-MD	202	11/52	1/52	0/0	0/0
28S-L	385	2/52	2/52	0/4	28/32
28S-LD	385	4/52	0/52	14/28	57/51

Primer sequences

Gene/pair	Forward	Reverse
16S-S	AATTTGGTTGGGACAACC	CTGTCAATGTGGTCTCTCTAGT
16S-SD	AATTTGGTTGGGACRAYY	CTGTCAATGTGRTCTCTCTARY
16S-M	AAGGACAAGAAGACCCTA	AATAGTTACACTAAGCCAAC
16S-MD	AAGGACAAGAAGACCCTA	AATAGTTACAYWRAGCCAAC
16S-L	TTAATTGAAGGCTAGTATGA	CAATGTGATCTCTCTAGT
16S-LD	TTAATTGAAGGYYRGTATGA	CAATGTGATCTCTCTARY
28S-S	GACGCGTGCACTTTCCGG	CAGAGGGCCTGCGACTCG
28S-SD	GACGCGTGCAMTTTCCRG	CAGAGGGCCTGCGACYHG
28S-M	CTGTAGCGTTAGGGACGTCGATCGAG	CGGACTCGTGCCAGTATTTAGCCTTAGA
28S-MD	CTGTAGCGTTAGGGACGTCGRTCGAV	CGGACTCGTGCCAGTRTTTAGCSTTRGA
28S-L	GGTGAGCACGGAAGTCTCG	CCGATTCCAGGGAGCTCGT
28S-LD	GGTGGGCACRGAAGYCTYG	CCGATTCCAGGGWRCTCGY

Binding coordinates, accession identifiers and per-species matches: panels/Nereididae/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix A8. Hesionidae

Canonical prefix: MiAphro-Hesionidae-. All sequences are 5′→3′. Covered and exact counts use the full target reference denominator. Strict and loose background values are WC/GT.

Gene/pair	bp	Covered	Exact	Strict bg	Loose bg
16S-S	188	28/53	2/53	1/7	1/7
16S-SD	188	37/53	1/53	1/43	32/44
16S-M	258	24/53	1/53	22/43	48/47
16S-MD	258	30/53	3/53	22/44	68/48
16S-L	362	28/53	2/53	2/8	7/27
16S-LD	362	37/53	3/53	2/60	50/69
28S-S	196	31/45	3/45	5/14	22/17
28S-SD	196	40/45	8/45	10/44	78/70
28S-M	252	28/45	6/45	1/4	17/7
28S-MD	252	35/45	7/45	2/8	47/11
28S-L	416	38/45	1/45	22/29	77/33
28S-LD	416	38/45	1/45	22/29	77/33

Primer sequences

Gene/pair	Forward	Reverse
16S-S	TGAACTAAGGTTTTAACGGC	AAACTCTATAGGGTCTTCTTGTCT
16S-SD	TGAATTACGGYTTHAACGGC	AAACTCAATAGGGTCTTCTTGTCY
16S-M	ACGGTTTTAACGGCCGCGGT	TCCCCAGTCGCCCCAACTAA
16S-MD	ACGGTTTTAACGGCYGCGGT	TCCCCAGTCGCCCCAACYAA
16S-L	TGAACTAAGGTTTTAACG	ATTAGCCTGTTATCCCTA
16S-LD	TGAATTACGGYTTHAACG	ATTAGCCTGTTAYCCCTR
28S-S	CACTTTCCGCGGGGAGAG	TGTGAACAGTCCTCCCCAGT
28S-SD	CACTTTCCGYSGRGAGAG	TGTGAACAGTCCTYCCMRGT
28S-M	GACGAGCGCACTTTCCGC	GAAGCACTGTGAACAGTCCT
28S-MD	GACGAGCGCACTTTCCRB	GAAGCACTGTRRRCAGTCCT
28S-L	GAGTCGGGTTGTTTGAGAATGCAGCCCA	CGAGCCATAGGACGGGCACGGC
28S-LD	GAGTCGGGTTGTTTGRGAATGCAGCCCA	CGAGCCATAGGACGGGCACGGC

Binding coordinates, accession identifiers and per-species matches: panels/Hesionidae/results.csv and species.csv in the repository. Ordinary/D pairs have identical endpoints within each design.

Appendix B. Conditional alternatives

Suffix 2 identifies a distinct alternative, not a mixture. Strict and close-relative gates are retained; loose-model counts are diagnostic rather than selection vetoes.

Target/gene/pair	Covered	Exact	Strict bg	Loose bg
Syllidae 16S S2	111/180	5/180	5/11	547/199
Syllidae 16S S2D	126/180	48/180	33/42	1007/601
Syllidae 16S M2	56/180	31/180	31/34	592/88
Syllidae 16S M2D	126/180	1/180	34/40	974/526
Syllidae 16S L2	44/180	14/180	3/5	993/806
Syllidae 16S L2D	91/180	50/180	12/15	2023/987
Nereididae 28S M2	15/52	11/52	0/12	3/83
Nereididae 28S M2D	19/52	1/52	6/12	248/91

Sequences (5′→3′)

Pair	Forward	Reverse
Syllidae S2	AAAGACAGAAAGACCCTA	TATTAGCCTGTTATCCCTATAGT
Syllidae S2D	AAAGACAAGAMGWCCCTR	TATTAGCCTGTTAYCCCTAHAGT
Syllidae M2	AGGTAGCATAATCATTTGC	TATTAGCCTGTTATCCCTACAG
Syllidae M2D	AGGTAGCACAATAAYTWGY	TATTAGCCTGTTATCCCTANAG
Syllidae L2	AAGGTAGCATAATCACTTGCCT	CCAACATCGAGGTGCCAATCA
Syllidae L2D	AAGGTAGCATAATMAYKTGCCT	CCAACATCGAGGTGCYAADCA
Nereididae M2	AGTAACGGCGAGTGAAGCGGGAA	CAAACAACCCGACTCTTGGGACAGCCG
Nereididae M2D	AGTAACGGCGAGTGAAGCGGGAR	CAAACAACCCGACTCYWAGGACAGCCR

Full names and coordinates are in conditional/results.csv; all eight pairs were independently re-scanned.
Copyright © 2026 Fei Xia. Repository copyright and third-party data terms apply.